Special Cases of Lossy Media

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Lecture Objective

By the end of this lecture, you will be prepared to:

- **Evaluate the parameters of plane waves propagating in certain important special cases of media.**

Important Special Cases

- Case 0: Lossless case
- Case 1: Low-loss non-magnetic dielectric
- Case 2: Good conductor

Lossless Case

- ε and μ are purely real; $\sigma = 0$.

$$\gamma = \sqrt{-\omega^2 \varepsilon' \mu'} = jk \qquad \Rightarrow \beta = k$$

$$\eta = \sqrt{\frac{\mu'}{\varepsilon'}} \qquad v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon' \mu'}}$$

Lossless Non-Magnetic Dielectric

$$\varepsilon = \varepsilon' = \varepsilon_r \varepsilon_0 \qquad \mu = \mu' = \mu_0$$

Relative permittivity a.k.a. dielectric constant

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \qquad v_p = \frac{\omega}{k} = \frac{c}{\sqrt{\varepsilon_r}}$$

Lossy Non-Magnetic Dielectric

$$\varepsilon = \varepsilon_r \varepsilon_0 (1 - j \tan \delta) \qquad \mu = \mu_0 \qquad \sigma = 0$$

Loss tangent a.k.a. dissipation factor (DF)

Relative permittivity a.k.a. dielectric constant (DK)

Low-Loss Non-Magnetic Dielectric

$$\varepsilon = \varepsilon_r \varepsilon_0 (1 - j \tan\delta) \qquad \mu = \mu_0 \qquad \sigma = 0$$

$$\tan\delta \ll 1$$

$$\gamma = j\frac{\omega}{c}\sqrt{\varepsilon_r}\sqrt{1 - j\tan\delta} \approx j\frac{\omega}{c}\sqrt{\varepsilon_r}\left(1 - j\frac{1}{2}\tan\delta\right)$$

$$\alpha = \frac{\omega}{2c}\sqrt{\varepsilon_r}\tan\delta \qquad \beta = \frac{\omega}{c}\sqrt{\varepsilon_r}$$

Low-Loss Non-Magnetic Dielectric Continued

$$\eta = \sqrt{\frac{\mu_0}{\varepsilon_r \varepsilon_0 (1 - j\tan\delta)}} \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1 + j\frac{1}{2}\tan\delta\right)$$

$$v_p = \frac{\omega}{\beta} \approx \frac{c}{\sqrt{\varepsilon_r}}$$

Non-Magnetic Good Conductor

$$\varepsilon = \varepsilon_0 \qquad \mu = \mu_0$$

$$\sigma >> \omega\varepsilon_0$$

$$\gamma = \sqrt{(\sigma + j\omega\varepsilon_0)j\omega\mu_0}$$

$$\approx \sqrt{j\omega\mu_0\sigma} \qquad = (1+j)\sqrt{\frac{\omega\mu_0\sigma}{2}}$$

Non-Magnetic Good Conductor Continued

$$\alpha = \beta = \sqrt{\frac{\omega\mu_0\sigma}{2}} = \sqrt{\pi f \mu_0 \sigma}$$

$$\eta = \sqrt{\frac{j\omega\mu_0}{(\sigma + j\omega\varepsilon_0)}} \qquad \approx \sqrt{\frac{j\omega\mu_0}{\sigma}} \qquad = (1+j)\sqrt{\frac{\pi f \mu_0}{\sigma}}$$

Non-Magnetic Good Conductor Continued

$$v_p = \frac{\omega}{\beta} = \sqrt{\frac{2\omega}{\mu_0 \sigma}} \xrightarrow[\sigma \to \infty]{} 0$$

Skin Depth of a Good Conductor

- For a good conductor:

$$\delta_s = \frac{1}{\alpha} \approx \frac{1}{\sqrt{\pi f \mu_0 \sigma}}$$

Question

- **What happens as the conductor becomes closer and closer to perfect ($\sigma \rightarrow \infty$)?**

Summary

- We have developed simple formulas for the parameters of plane waves propagating in low loss dielectric materials and in good conductors.

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